Design and Implementation of High-performance Lock Preamplifier for High-Field Nuclear Magnetic Resonance Spectrometers

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Abstract:*In this paper, based on the field-frequency lock technology commonly used in the current commercial nuclear magnetic resonance spectrometers to maintain the magnetic field homogeneity and the theory of weak signal detection, a lock preamplifier at the front-end of lock channel is designed and implemented. The preamplifier consists of broadband amplifier, RF switch and attenuation circuits. In the 20-200 MHz the fabricated preamplifier exhibits a gain of 53 dB and in-band fluctuation of 1 dB. The whole noise figure of receiving channel is about 1.5 dB. The delay of RF switch is less than 220 ns. For the transmitter channel the total attenuation range is 30 dB with a 2.0 dB minimum step size. Compared with the lock preamplifier of the commercial spectrometers, the developed preamplifier shows a lower noise figure, a smaller in-band fluctuation and a shorter delay of the switch. Therefore the preamplifier can be applied to nuclear magnetic resonance spectrometers from 200 MHz to 900 MHz.*
Keywords:*NMR, lock preamplifier, field-frequency lock technology.*

I. INTRODUCTION

Nuclear magnetic resonance (NMR) spectrometers are widely applied in scientific research and clinical applications. In order to obtain high resolution of signal, current commercial spectrometers tend to high-field development [1], which requires high performance for the structural parts. High-resolution signals need the homogeneous magnetic field, so the field-frequency lock technology and the corresponding parts of spectrometer play a key role in the shimming process [2-3]. Therefore the performance of lock preamplifier in the lock channel is essential to lock signal transmitting and receiving, determining the shimming quality and sensitivity of signal detected [4]. Now the field-frequency lock technology is widely used in the current NMR spectrometers. In this paper the requirements of preamplifier performance for the lock channel of spectrometer are analyzed, as well as the design and implement of preamplifier are completed. The amplification module based on the theory of low-noise broadband amplifier was designed [5]. In the transmitter mode there are attenuation circuits where transmitter power can be attenuated from 0 dB to 30 dB with a 2.0 dB minimum step size. The results show that in 20-200 MHz frequency range, the lock preamplifier achieves a gain of about 54 dB with in-band fluctuating of 1dB, and a noise figure of about 1.5 dB. The delay of the radio frequency (RF) switch is less than 220 ns and the isolation is more than 65 dB. With a low noise figure of the broadband, high switching speed and detection sensitivity, the preamplifier could be applicable to NMR spectrometers from 200 MHz to 900 MHz.

II. THEORETICAL ANALYSIS

A. Field-frequency lock system

NMR occurs, when:

$$2\pi f_o = \gamma B_o \tag{1}$$

f_o is resonance frequency, B_o is magnetic field intensity, γ is gyromagnetic radio. If either f_o or B_o is not stable, the spin system will deviate from the resonance point. The sensitivity and resolution of detection signals will reduce, even the spectral line will disappear completely.

Eq. (1) can be written as:

$$f_o / B_o = \gamma / 2\pi \tag{2}$$

Eq. (2) indicates that the radio frequency and the magnetic field intensity must maintain a constant ratio to satisfy the stability condition. In general radio frequency can maintain highly stability. The magnetic field is not always constant because of the drift of superconducting current and the influence of external factors. Therefore in the field-frequency lock systems the amplified deviation signals of the magnetic field are used to correct the changes of the magnetic field intensity, while the radio frequency is fixed.

In Figure 1, a NMR dispersion signal as a reference axis is shown [6]. The amplitude of the dispersion signal is zero when $\omega_0=\gamma H_0$.When the magnetic field drifts, H_0 has a offset ΔH_0 and the amplitude of corresponding dispersion signal will offset to zero, resulting in a positive or negative voltage. After the positive or negative voltage is detected, a compensating current with appropriate amplitude and direction is sent to the shimming coil. The method will keep the magnetic field stable. By this way it can ensure the magnetic field intensity is still H_0, returning to the resonance condition [7]. It is known as field-frequency lock technology and

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makes the homogeneity of magnetic field more than 10^{-10}.

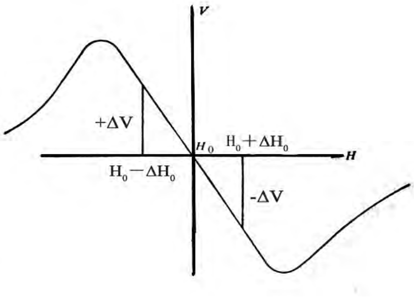


Fig.1. Principle of the field-frequency lock

A field-frequency lock system is shown in figure 2. The lock RF pluses from the lock transceiver are generated by the reference signals from the clock synchronization and the offset signals from the lock transceiver controller. In the transmitter mode, the RF pluses are applied to the switch of the lock preamplifier, at last to the lock coil in the probe. In the receiving mode the lock signals are detected by the coil and amplified by the broadband amplifier section of the lock preamplifier. After that, the signals are sent to the lock transceiver, where they are mixed with the local oscillator and phase detected into two orthogonal signals. One output of the orthogonal signals is applied to the lock transceiver controller to adjust the offset frequency. The other output of the orthogonal signals is applied to the shim systems to keep the magnetic field constant.

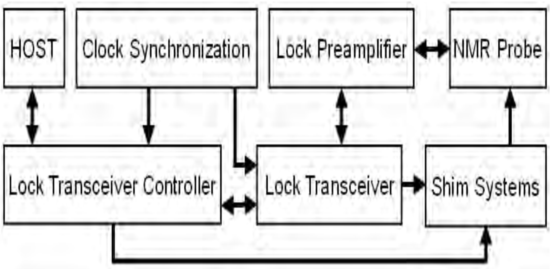


Fig.2. Block diagram of the field-frequency lock system

B. Lock preamplifier structure and performance requirements

As a front-end part of the lock channel, the lock preamplifier consists of broadband low noise amplifier, RF switch circuits and attenuation circuits of transmitter power. In the transmitter mode, RF pulses, from the lock transmitter channel, pass through the attenuation circuits, and enter the lock coil of probe to excite sample to generate resonance signals. In the receiving mode, the lock signals from the probe which are free induction decay (FID) signals, pass through RF switch circuits, and then are amplified by the broadband preamplifier, and finally enter the lock receiving channel.

Modern commercial high-field spectrometers commonly adopt heteronuclear-internal lock system where nuclear observed is different from nuclear locked. Deuterium with lower resonance frequency is usually used as a locked sample. For example, the resonance frequency of deuterium is 76.75 MHz for 500 MHz spectrometer and 138.15 MHz for 900 MHz spectrometer. In this paper, the lock preamplifier is designed to work from 20 MHz to 200 MHz and can be applied to 200-900 MHz spectrometers.

In the experimental process, the entire lock system frequently detect the resonance frequency of the deuterium solvent, so it requires the preamplifier has a short delay of the switch from transmitter mode to receiving mode.

As the lock signals received from the probe are micro-volt level, it requires the preamplifier not only has a higher gain and a lower noise figure, but also a good dynamic range that decides detection resolution.

According to the RF design principle, the noise figure of the multi-stage small signal amplifiers mainly depends on the first and the second stage. The formula of noise figure is as follows:

$$F = F_1 + \frac{F_2 - 1}{G_1} + \dots + \frac{F_N - 1}{G_1 G_2 \dots G_{N-1}} \tag{3}$$

In a weak signal detection system, a lower limit of dynamic range is usually required, otherwise it is impossible to extract the useful signal from the noise [8-9]. The smallest signal detection equation is as follows [10]:

$$E_i = \sqrt{4kTR_sB \cdot F \cdot SNR_o} \tag{4}$$

where k is Boltzmann constant, k=1.38×10-23 J/K; T is the absolute temperature of the load; Rs is the impedance of the load; B is the effective bandwidth for the noise; F is the noise figure and SNR_0 is the signal-to-noise ratio of output. Eq. (4) indicates that the lower noise figure is, the smaller E_i is and the higher detection resolution is. According to the lock channel and the lock coil parameters in the probe, when the noise figure of preamplifier is 1.5 dB, the small signal that is less than 5 uV (-93 dBm) can be detected.

III. LOCK PREAMPLIFIER DESIGN AND SIMULATION

Figure 3 shows the block diagram of lock preamplifier designed. The schematic of the RF transceiver is shown in Figure 4. A GaAs FET SPDT chip is applied in RF switch circuit. Gating control of the switch is excited by the lock synch signal from the lock transceiver controller. When the switch turns to the transmitter mode, the RF pulses pass through a 30 dB switchable attenuator and are directed to the probe. The attenuator is a GaAs FET 4-bit digital attenuator with a 2.0 dB minimum step size and a

30 dB total attenuation range. Attenuation value is controlled by logic gates. Protection circuit is located between the probe and the RF switch. If the input power of the lock transmitter is too high, the shunt diodes will conduct to prevent the lock coil of the probe from damage.

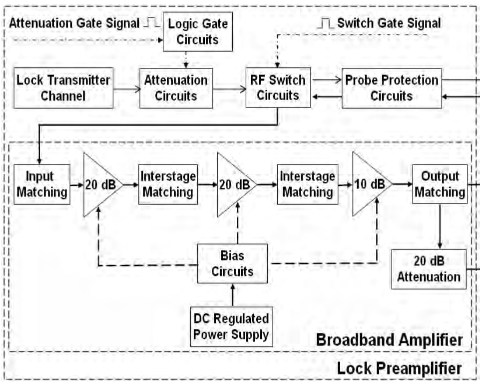


Fig.3. Block diagram of the lock preamplifier

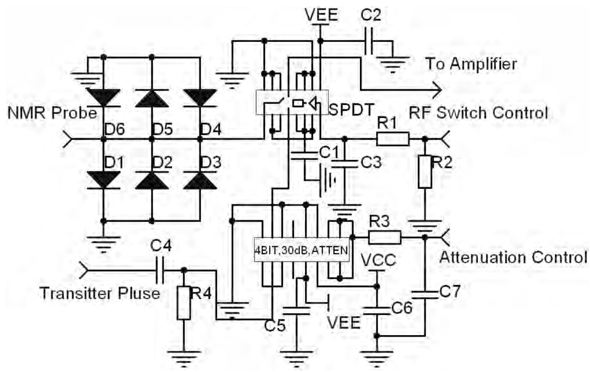


Fig.4. Schematic of the RF transceiver

The broadband amplifier is composed of three-stage amplifiers, as shown in Figure 5. The first-stage is designed based on the minimum noise figure theory. The second-stage and third-stage are designed based on the maximum power gain theory. Therefore the amplifier can satisfy noise and gain requirements.

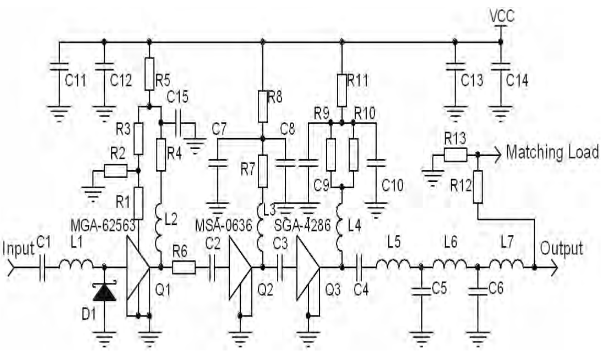


Fig.5. Schematic of the low noise broadband amplifier

The first-stage amplifier used in the circuits is MGA-62563, a high performance GaAs Monolithic Microwave Integrated Circuit (MMIC) amplifier with Avago Technologies E-pHEMT process. The amplifier operates at V_d=3V and I_d=25 mA to deliver the gain of 21 dB approximately in 20-200 MHz. The input matching network consisting of C1, L1 and D1 is designed for optimum noise figure, since the noise figure of the whole amplifier is mainly decided by the first stage. At the input circuit a series air core inductor (L1) is placed to match the input of the device for low Voltage Standing Wave Ratio (VSWR). Shunt diode D1 protects the preamplifier against higher power input. R6 is used to reduce the excessive gain of the active device in the whole frequency range, thereby improving stability [11].

MSA0686 is used as the second-stage amplifier, which is a high performance silicon bipolar MMIC housed in a cost effective. R7 and R8 are parts of the bias network, which are used to stabilize the drain voltage at 3.5 V and the drain current at 20 mA. DC blocking capacitors (C2 and C3) are used at the input and output of the MMIC to isolate the device from the preceding stage and subsequent stage respectively.

SGA-4286 is used in the third-stage amplifier because its Darlington configuration featuring one micron emitters provides high F_T and excellent thermal performance at a low supply voltage. With the bias of V_d=3.0 V and I_d=43.5 mA, the gain is about 14 dB in 20-200 MHz. To attain lower in-band fluctuation and output VSWR, a compact output matching circuit with low-pass filtering can be achieved by L5, L6, L7, C5 and C6.The attenuation of output network consisting of R12 and R13 is used to enhance the stability factor of the broadband amplifier.

The circuits of the broadband amplifier were simulated with ADS2005A. The analysis of S-parameters and noise figures were carried out after the establishment of a good DC-bias simulation. After the layout of the initial circuits was formed, the broadband amplifiers were simulated again, including the actual component models, parasitic packages and discontinuities.

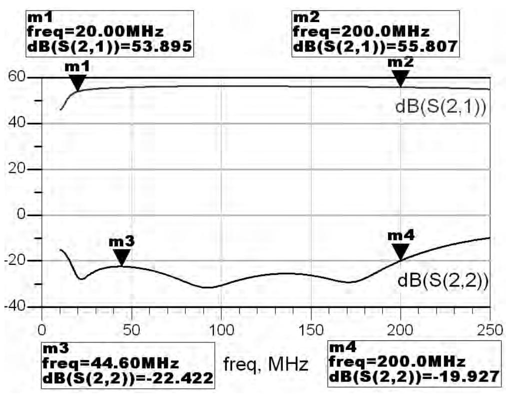


Fig.6. Simulated small signal S-parameters of the amplifier

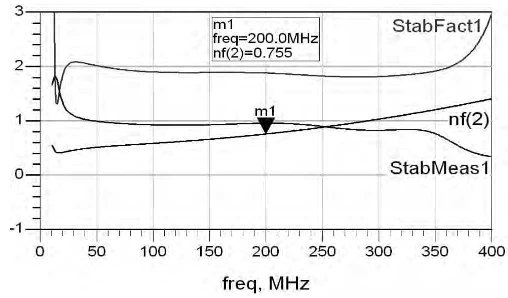


Fig.7. Simulated noise figure and stability factor of the amplifier

The simulation results are shown in Figure 6 and Figure 7, which indicates that the amplifier is stability in the frequent from 10 MHz to 400 MHz. In 20-200 MHz, the gain is about 54 dB; the noise figure is less than 0.8 dB, the In-band fluctuation is less than 2 dB and the output VSWR is less than 1.2.

IV. PCB DESIGN AND TEST RESULTS

The printed-circuit board (PCB) was fabricated with FR4 substrate. All the S-parameters of the preamplifier, input and output VSWR, 1 dB compression point were measured by Agilent network analyzer E8362B. Since the maximum input power and minimum output power of the network analyzer was 20 dBm and -27dBm respectively, the first-port of the network analyzer was connected with a 30 dB attenuator and the output power was set to -20 dBm in the test. The measured results shown in Figure 8 indicates: in the 20-200 MHz, the gain is about 54 dB; band flatness is less than 0.7 dB and output VSWR is less than 1.25. 1dB compression point, shown in Figure 9, at 76MHz is 13 dBm. All the results are in good accordance with design specifications.

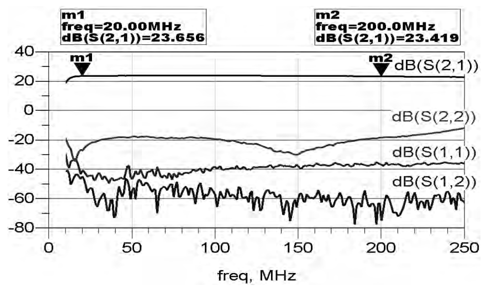


Fig.8. Measured small signal S-parameters of the preamplifier

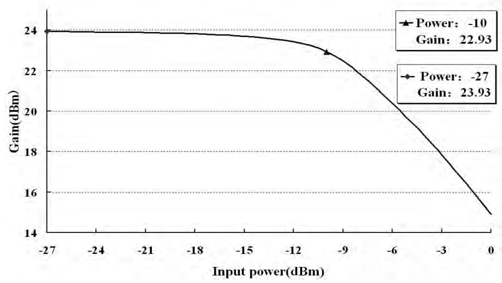


Fig.9. Measured 1dB compression point of the preamplifier

The noise figure in the 20-200MHz was measured by noise analyzer HP-8970A. As shown in Figure 10, the noise figure in-band is about 1.5dB. In the separate amplification module, the noise figure of broadband is 0.8 dB, which is in accordance with simulation results. Therefore the rest of noise figure is contributed by the RF switch.

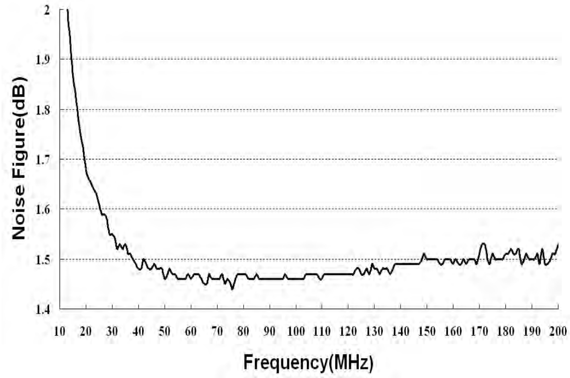
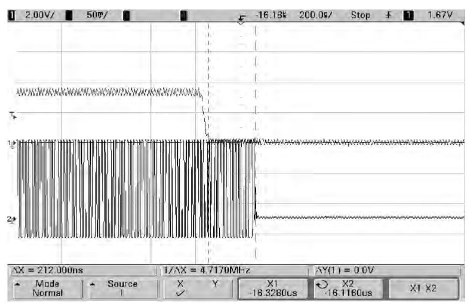
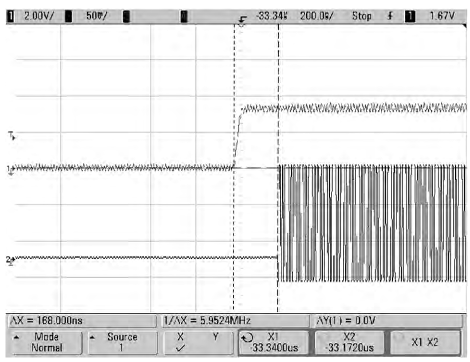


Fig.10. Measured noise figure of the preamplifier

The delay of the RF switch circuits was observed with Agilent oscilloscope DSO6104. Figure 11 (a) and (b) show that the delay of RF switch is less than 220 ns. All the performances are shown in Table 1.



(a)



(b)

Fig.11. Measured delay of RF switch (a) from transmitter mode to receiving mode and (b) from receiving mode to transmitter mode

Table1. The overall performances of the lock preamplifier

	Result	Unit
Frequency	20-200	MHz
Gain	53.5±0.5	dB
In-band Fluctuation	< 1	dB
Noise Figure	1.5±0.1	dB
P_{out} 1dB Compression Point	>12	dBm
Delay of Switch	<220	ns

V. CONCLUSION

In this paper, based on the field-frequency lock technology, which is commonly used in the current high-field NMR spectrometers, the performance requirements of the lock preamplifier for the lock channel are analyzed. The design and implementation of the lock preamplifier have been completed. The results show that the preamplifier has a broad bandwidth (from 20MHz to 200MHz), a flat gain, a good output return losses, short delay of RF switch and a low noise figure in the whole receiving channel. The preamplifier can be applied to 200-900MHz NMR spectrometers.

VI. ACKNOELEDGEMENT

This work is supported by National Key Project of Scientific and Technical Supporting Programs Funded by Ministry of Science & Technology of China (2006BAK03A222).

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